THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In deserts, nights are cold and days are hot because sand: 1

(a) heats and cools quickly **(b)** heats slowly
(c) stores heat **(d)** reflects all

2. A chimney is made tall so that: 1

(a) hot gases rise and draw fresh air **(b)** bricks cool
(c) rain cannot enter **(d)** smoke is hidden

3. A wall painted white keeps the room cooler than a wall painted black because: 1

(a) white reflects most radiation **(b)** black reflects
(c) white conducts **(d)** white absorbs

4. Water in tanks near the sea stays warm at night because: 1

(a) water is heated by the moon **(b)** water cools slowly
(c) water is hot **(d)** wind is warm

5. A house in a hot area should have windows high up because: 1

(a) dust is kept out **(b)** hot air can escape
(c) light enters **(d)** cold air escapes

6. At the same temperature, a metal chair feels colder than a wooden chair because: 1

(a) metal is colder
(b) wood is hotter
(c) metal conducts heat away from the body faster
(d) wood is a conductor

7. Clouds at night keep the ground warmer because: 1

(a) they cool the ground **(b)** they heat the sun
(c) they carry light **(d)** they reduce loss of radiation

8. The temperature near the ceiling of a closed room with a heater is: 1

(a) lower **(b)** same
(c) higher than near the floor **(d)** zero

9. **Assertion (A):** Convection takes place in liquids and gases. 1
Reason (R): Conduction takes place in solids.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A sea breeze blows from the sea to the land during the day. 1
Reason (R): During the day, the sea water gets hotter than the land.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are cooking vessels sometimes blackened at the bottom? 2

12. Why does a cold glass bottle feel colder than a cold plastic bottle at the same temperature? 2

13. Explain why desert nights are cold though days are hot. 2

14. Why do fire-fighters wear shiny suits? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Design an experiment to prove that heat travels through a vacuum. 3

16. Why do firemen cut a wall of vegetation as a firebreak? 3

17. Explain why electric kettles have heating elements near the bottom. 3

OR

Explain the role of land and sea breezes on the climate of coastal cities. 3

18. Explain why clothes dry faster in sunshine than in the shade. 3

OR

Why does a closed car parked in the sun become hot inside? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meera lives in Chennai near the sea. On a hot afternoon she feels a cool breeze coming from the sea to the land. At night, the breeze flows in the opposite direction. 5
(i) Which heats faster during the day, land or sea?
(ii) What is the afternoon breeze called?
(iii) What is the night-time breeze called?
(iv) Explain why the breeze blows from the land to the sea at night.

20. Explain the heating of the Earth by the Sun and the role of the atmosphere, land and sea in producing winds. 5

OR

Explain the working of a double-glazed window in cold climates. 5

END OF QUESTION PAPER — Total: 40 Marks